

# SEOJUNG MIN

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Work Authorization: EAD approved; no sponsorship required

## RESEARCH INTERESTS

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My research aims to enable robots to understand and act through physical contact by transforming tactile signals into task-relevant representations for manipulation.

- **Contact-rich robotic manipulation:** Inferring contact states, environmental constraints, and tool-mediated interaction from tactile sensing and robot proprioception
- **Tactile perception and representation learning:** Estimating force, geometry, and material properties from vision-based and magnetic tactile sensors under real-world variation
- **Sensor-integrated robotic systems:** Building experimental platforms for tactile data collection, calibration, validation, and robot interaction

## EDUCATION

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**Korea Advanced Institute of Science and Technology** 2022 - present

Ph.D. student in Mechanical Engineering, Advisor: Prof. Jung Kim

Research Focus: *Contact-rich manipulation and tactile perception for real-world robotic systems*

**Korea Advanced Institute of Science and Technology** 2013 - 2015

M.S. in Mechanical Engineering, Advisor: Prof. Jung Kim

Thesis: *Inertial sensor-based ambulatory motion sensing for inverse dynamics analysis of human motion*

**Korea Advanced Institute of Science and Technology** 2008 - 2013

B.S. in Mechanical Engineering

## SKILLS

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|------------------------|-------------------------------------------------------------------------------------|
| <b>Robotics</b>        | ROS, MoveIt, UR5e, Kinova, Robotiq gripper, MuJoCo, Unity                           |
| <b>Programming</b>     | Python, MATLAB, C/C++, Java, LabVIEW                                                |
| <b>Tactile Sensing</b> | Vision-based sensors (GelSight Mini, DIGIT, GelSlim 3.0), Magnetic sensors (eFlesh) |
| <b>Hardware</b>        | PCB design (KiCad, EasyEDA), Microcontrollers (Teensy, Arduino, STM32)              |
| <b>Prototyping</b>     | SolidWorks, 3D Printing, Silicone Fabrication, Laser Cutting                        |
| <b>Other Systems</b>   | Mixed Reality (HoloLens 2), Motion Capture (OptiTrack), IMU, sEMG                   |

## PROFESSIONAL EXPERIENCE

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**Agency for Defense Development** 2017 - 2019

Researcher, Modeling & Simulation Technology Lab.

*Daejeon, S.Korea*

- Developed reusable and modular software components for naval weapon systems
- Conducted system-level war simulations to validate reliability and reusability of developed models

**Samsung Electronics** 2016 - 2017

Engineer, Washing Machine Expert Lab.

*Suwon, S.Korea*

- Designed PI velocity controllers for motors and implemented wash algorithms
- Optimized algorithm parameters based on extensive experimental testing

## PROJECTS

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### **Extrinsic Contact Inference (Ph.D. Research)**

2024 - present

- Built an end-to-end robotic manipulation system integrating dual GelSight Minis, a Robotiq 2F-85 gripper, and a UR5e robot
- Developed online algorithm to localize a moving extrinsic contact point from tactile deformation and robot proprioception
- Collected synchronized tactile/force dataset for learning-based contact force inference
- Designed controlled contact-rich manipulation experiments for tool-mediated interaction and real-world robotic validation

### **Tactile Sensor-Based Contact Property Estimation**

2025 - present

- Fabricated and calibrated open-source magnetic tactile sensors
- Collected tactile roughness datasets using DIGIT sensors and a Kinova robot arm across multiple materials and sensing conditions
- Proposed and validated a ranking-based tactile representation learning framework for robust roughness estimation under sensor and material domain shifts

### **Robotic Grasp Perception Using Vision-Based Tactile Sensors**

2022 - 2023

- Fabricated and calibrated open-source GelSlim 3.0 tactile sensors
- Built a real-time pose and contact estimation pipeline and conducted controlled experiments with OptiTrack ground-truth tracking

### **Wearable Haptics & Proprioception Enhancement**

2022 - present

- Designed daisy-chain modular vibrotactile feedback hardware (FPCB, IMU interface, Teensy controller)
- Developed IMU-based motion-intent detection algorithms for vibrotactile proprioception augmentation

### **Enhancing AR Interaction with Vibrotactile Feedback (HoloLens 2)**

2022 - 2023

- Built an AR interaction prototype combining hand tracking with fingertip vibrotactile feedback
- Designed a cube-pushing click task comparing vision-only, audio, and haptic feedback
- Evaluated the prototype in a cube-pushing task, where haptic feedback improved task efficiency

### **Earlier Work (Human Motion Tracking, EMG, Exoskeleton, Sensors)**

2013 - 2015

- IMU-based inverse dynamics analysis for human motion
- Inertial sensor-based wrist-worn activity tracking system (KAIST & H3 System Co.Ltd.)
- Hand gesture recognition for sEMG interfaces (KAIST & Samsung Electronics)
- Bio-signal-based elbow force estimation models for exoskeleton applications (KAIST & LG Electronics)
- Muscle-supporting exoskeleton system with sEMG-based control

## PUBLICATIONS

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### ***JOURNAL ARTICLES***

#### **In-hand object classification and pose estimation with sim-to-real tactile transfer for robotic manipulation**

Yang, Sanghoon, Won Dong Kim, Hyunkyu Park, **Seojung Min**, Hyonyoung Han, and Jung Kim  
*IEEE Robotics and Automation Letters* 9.1 (2023)

### ***CONFERENCE PAPERS***

#### **Real-Time Evaluation and Few-Shot Hardware Adaptation of a Vision-Based Tactile Sensor**

Juhee Park, **Seojung Min**, and Jung Kim  
*2026 26th International Conference on Control, Automation and Systems (ICCAS)*, Accepted.

#### **Geometric Reconstruction of Extrinsic Contact Trajectories using Tactile Sensing and Proprioception for Tool Manipulation**

**Seojung Min**, Yoonjin Kim, Jeong-Jung Kim, and Jung Kim  
*2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Accepted.

#### **Ranking-Based Learning for Robust Tactile Roughness Perception under Domain Shift**

**Seojung Min**, Juhee Park, Gunhee Park, and Jung Kim  
*2026 23rd International Conference on Ubiquitous Robots (UR), IEEE, 2026*, Accepted.

#### **Preliminary Study of Portable Vibrotactile and Electrical Stimulation System for Knee Proprioception Enhancement**

Park, Sungbin, **Seojung Min**, Junhwan Choi, Min Jin Yang, Dayoung Yeom, Wonseok Shin, Chaeree Park, and Jung Kim  
*2023 20th International Conference on Ubiquitous Robots (UR), IEEE, 2023*

#### **Inertial sensor based inverse dynamics analysis of human motions**

**Seojung Min** and Jung Kim  
*2015 IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*

#### **Personalized protocol to select usable movements for myoelectric pattern recognition**

Na, Youngjin, Sangjoon J. Kim, **Seojung Min**, Changmok Choi, Sang Joon Kim, Sungho Jo, and Jung Kim  
*6th European Conference of the International Federation for Medical and Biological Engineering (2015)*

#### **Investigation of a tolerable time delay in SEMG-based elbow assistive device**

Chang, Handdeut, Gwang Min Gu, **Seojung Min**, Hyosang Lee, and Jung Kim  
*2014 Proceedings of the SICE Annual Conference (SICE), IEEE*

### ***KOREAN PATENTS***

#### **Architecture design of base system model using engagement simulation**

**Seojung Min**, Sangjin Lee, Juhye Park, and Hyunsik Oh  
KR Patent Application No. 10-2018-0043505, filed Apr. 2018, granted Nov. 2018

#### **Method of managing and sharing object data based on spatial data for simulation of object-based engagements**

**Seojung Min**, Sangjin Lee, Juhye Park, and Hyunsik Oh  
KR Patent Application No. 10-2018-0046248, filed Sep. 2018, granted Sep. 2018

## TEACHING

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### Teaching Assistant: Coursework

KAIST

- Special Topics in Mechanical Engineering (ME493) Fall 2024, Spring 2025
- Mechatronics System Design (ME203) Fall 2015, 2022, 2023
- Applied Electronics (ME207) Spring 2015, 2023
- Engineering Design (ME340) Spring 2022

### Mentorship: Undergraduate Research Program

KAIST

- *Development of sEMG interface and haptic device for metaverse applications* Summer 2022